

Unit 5, Lesson 8: Determining Distances

Benchmarks

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

MA.6.GR.1.3 Solve mathematical and real-world problems by plotting points on a coordinate plane, including finding the perimeter or area of a rectangle.

Learning Targets

- ☐ I can find distances between ordered pairs.
- ☐ I can solve a problem by plotting points on a coordinate plane and determining distances.

5.8.1 Warm-Up: Superpowers

1. If you could have a superpower, what would it be and why?

2. Think of a superhero whose powers relate to mathematics.
 - a. Which superhero do you have in mind?

 - b. Explain how his or her superpowers relate to mathematics.

5.8.2 Guided Instruction: The Flash vs. Superman

The Flash challenges Superman to a race. Who do you think will win?

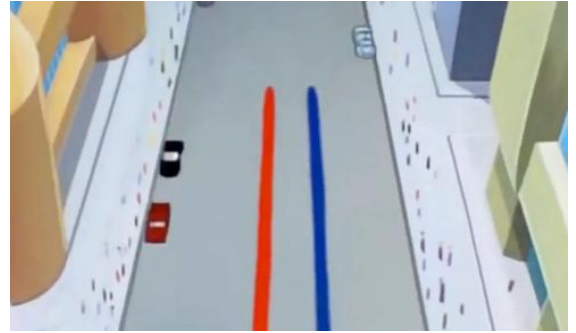


Without watching the race, answer the following.

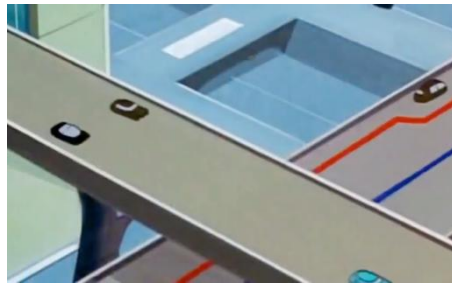
1. How can we find out the distance they covered?
2. How can we find out who won?

Unfortunately, in the movie, *Justice League*, we were not able to see who wins. However, the Flash and Superman did race each other in the cartoon and comics version.

They raced through cities...



and on highways...



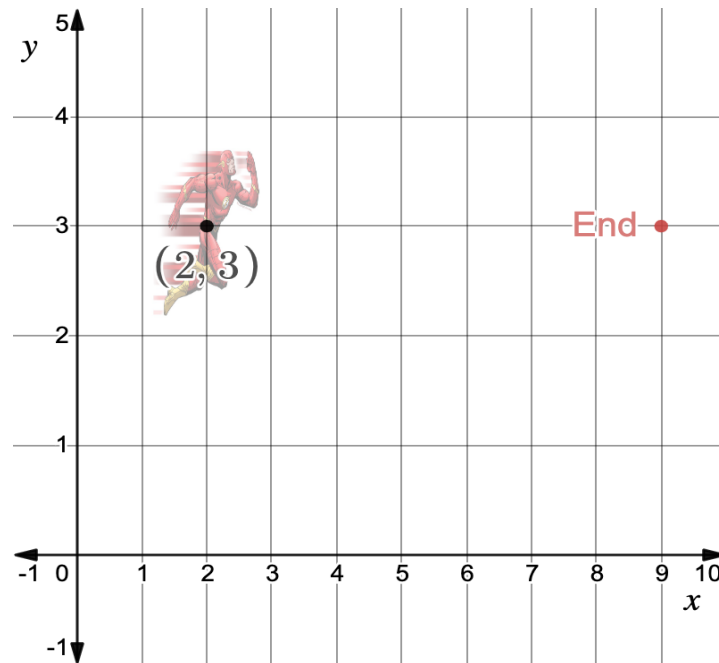
and through all kinds of conditions.



3. What do you see from the comic that is related to determining distance?

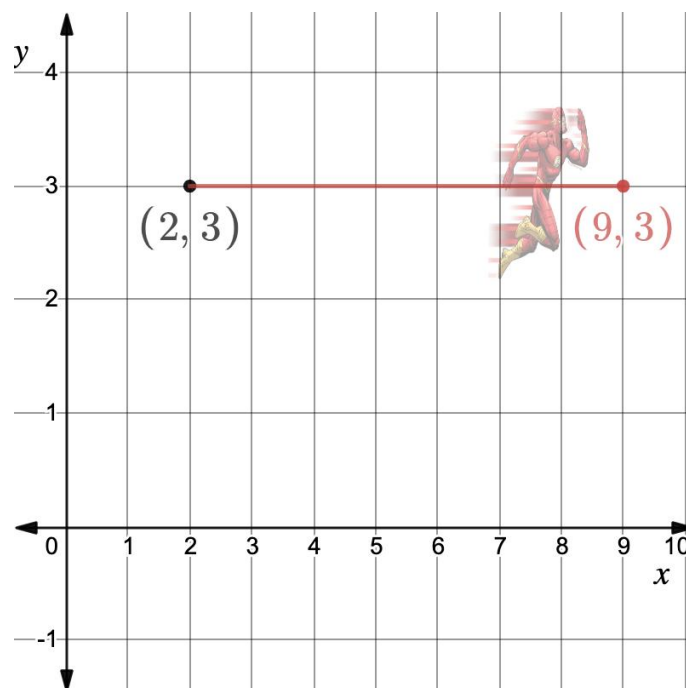
5.8.3 Exploration: Looking at a Trajectory

1. The following graph shows the Flash running from point $(2, 3)$.



What are the coordinates for where he ends?

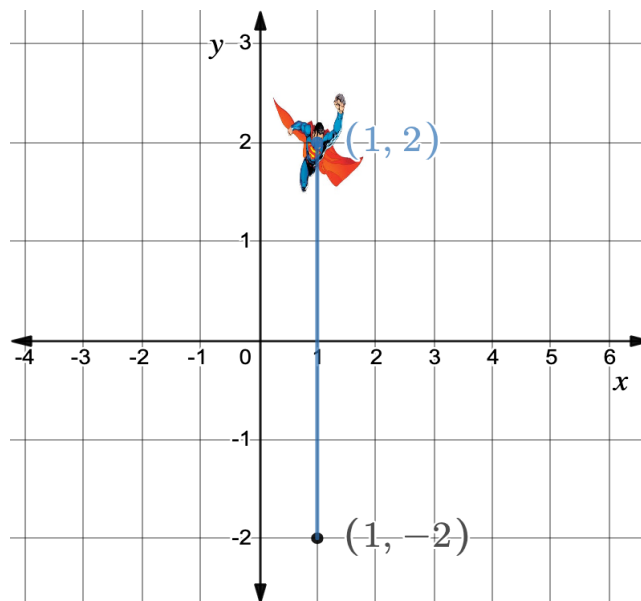
2. Consider the following graph.



- a. If each gridline on the x -axis represents 1 mile traveled, how many miles did the Flash travel?
- b. What is the distance between the starting and ending points for the Flash?
- c. What do you notice between the coordinates of the starting point in relation to the coordinates at the end point?
3. When finding the horizontal distance between two points that lie on the same line, if the points are on the same side of the y -axis, find the

sum
difference

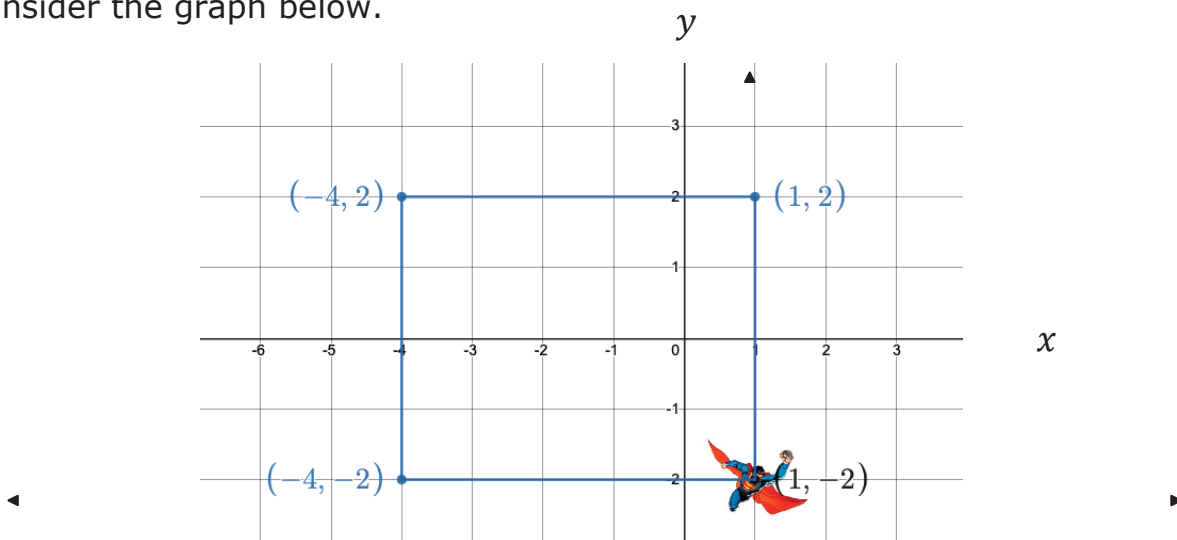
 of the absolute values between the x -values of the starting and ending points.
4. In the graph below, Superman starts flying at point $(1, -2)$. The y -axis represents the miles Superman travels.



- a. How many miles does he need to travel to get to point $(1, 2)$?
- b. How do the coordinates of the starting point relate to the coordinates at the end point?
5. When finding the vertical distance between two points that lie on the same line, if the points are on opposite sides of the x -axis, find the

sum
difference

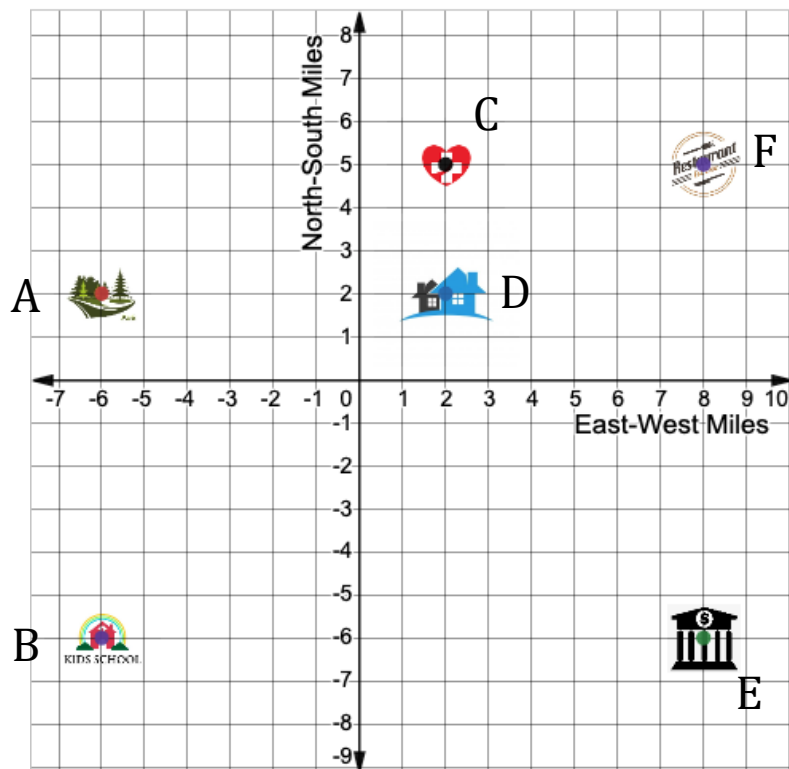
 of the absolute values between the y -values of the starting and ending points.
6. Consider the graph below.



- a. The blue line shows the trajectory Superman followed starting from $(1, -2)$. Assuming the x - and y -axes represent miles traveled, explain how to determine the distance.
- b. What is the total distance traveled by Superman?

5.8.4 Guided Practice: Town Map on a Coordinate Plane

1. Consider the following graph.



- a. What are the coordinates of the following places in town?

Park (A): _____

School (B): _____

Hospital (C): _____

House (D): _____

Bank (E): _____

Restaurant (F): _____

- b. Find the following distances.

The distance between the park at $(-6, 2)$ and the school at $(-6, -6)$ is _____ miles.

The distance between the hospital at $(2, 5)$ and the house at $(2, 2)$ is _____ miles.

The distance between the bank at $(8, -6)$ and the school at $(-6, -6)$ is _____ miles.

The distance between the restaurant at $(8, 5)$ and the hospital at $(2, 5)$ is _____ miles.

2. Loki leaves his house and heads towards the park. Then, he goes to school.

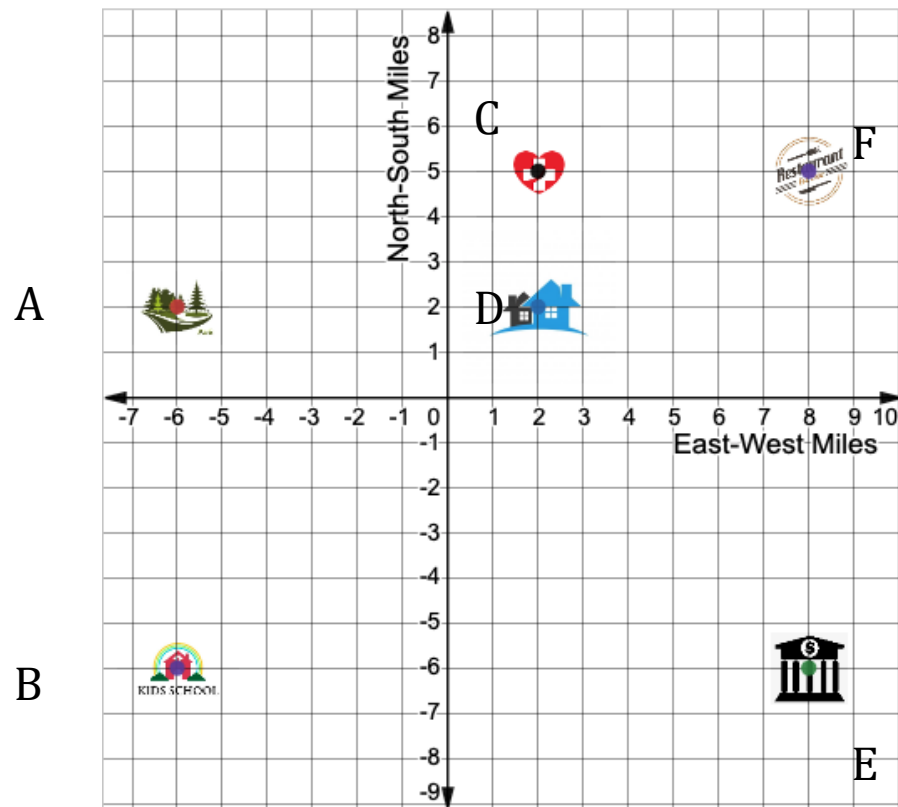
- a. What distance has Loki travelled?

- b. Loki gets out of school and goes to the bank to make a few transactions. Then, he heads toward the restaurant to have dinner. What distance has Loki traveled so far?

- c. Before going home, Loki needs to stop at the hospital to visit a friend. After leaving the hospital, he finally goes home.

What distance did Loki travel altogether?

3. The city commissioner will build a stadium so that it forms a square with the house, the park, and the school.

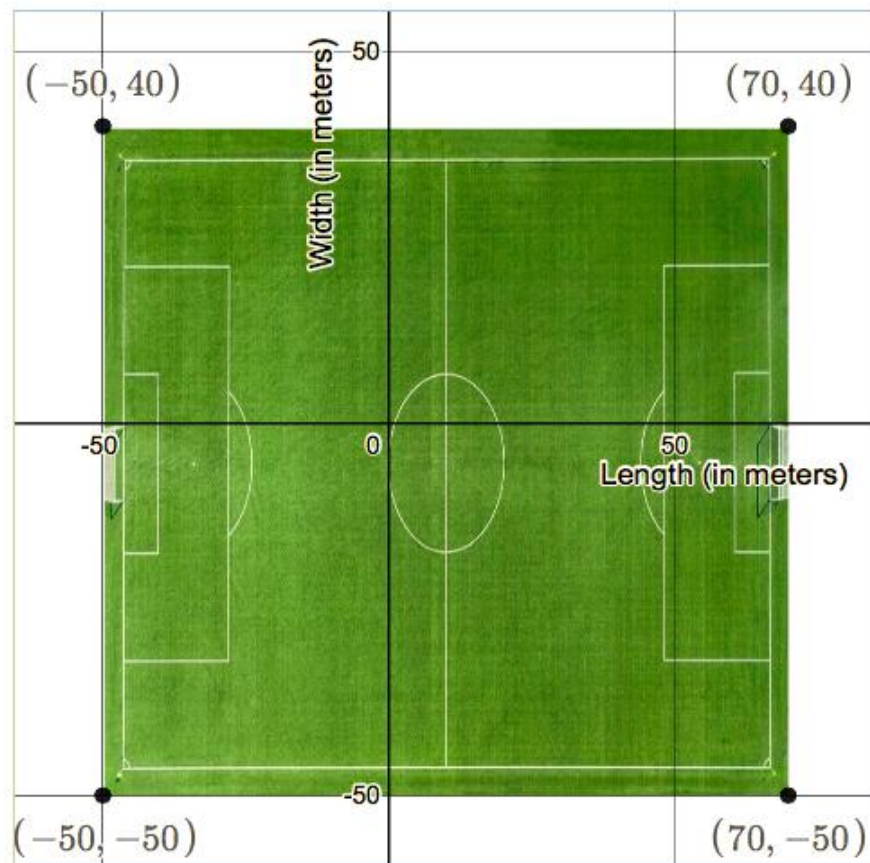


Explain where the stadium should be located.

4. Now that the stadium is built, what is the area and the perimeter formed by the stadium, the house, the park, and the school?
- a. Perimeter: _____ miles (mi.)
- b. Area: _____ (mi.²)

5.8.5 Your Turn!

1. Consider the soccer field on the coordinate plane.



- a. Find the dimensions of the soccer field.
- Length (horizontal): _____ meters (m)
- Width (vertical): _____ m
- b. What is the perimeter, in meters, of the soccer field?
- c. What is the area, in square meters, of the soccer field?

5.8.6 Wrap-Up: Ticket Out the Door

1. Zander is a competitive dart player. The graph shows a record of his last three darts.
 - a. Plot the dart, Dart Z, where Zander should throw his next dart so that it forms a rectangle with the other three on the plane.
 - b. What is the perimeter of the rectangle formed by the darts?
 - c. What is the area of the rectangle formed by the darts?
 - d. Zander will take back Darts P and Q. He wants to throw them so that they land on two points, forming a square with Darts T and Z. Where should the Darts P and Q land?

